SIEG GABRIEL G. SOLA

sieggabrielsola12@gmail.com | www.linkedin.com/in/sieg-gabriel-sola | https://solasieggabriel.github.io/portfolio/

PROFILE

Cum Laude BS Mathematics and DOST-SEI Scholar graduate with hands-on experience building hybrid variational quantum machine learning architectures and statistical workflows in Python and R. Seeking to leverage statistics and data analytics skills in industry-leading technology and research roles.

EDUCATION

Southern Luzon State University **Graduated: May 2026**

Bachelor of Science in Mathematics

- GWA: ~1.44 *Cum Laude*
- DOST-SEI scholar

EXPERIENCES

DOST - Advanced Science and Technology Institute | *March 2026 - April 2026*

Quantum Image Representation Intern

- Developed and benchmarked quantum encoding strategies in Python for hybrid quantum machine learning algorithm.
- Co-authored a comprehensive study on model architectures and performance metrics, enhancing understanding of quantum applications.
- Designed and ran statistical validation procedures in Python and R to assess the outputs of 144 stochastic model configurations using non-parametric tests, post-hoc comparisons, and correlation analysis.

LGU - Rizal, Laguna | *June 2025 - August 2025*

Accounting Intern

- Verified financial records by cross-checking expense reports, receipts, and payroll computations against offical documents whole coordinating with municipal departments to resolve discrepancies and obtain approvals
- Digitized 11 Barangays' financial data from January to June 2025 into Microsoft Excel, creating a searchable database for the municipal accounting office.

Photomath Inc. | *August 2020 - January 2024*

Freelance Math Expert

- Produced 2800+ of step-by-step technical solutions in LaTeX for a global student user base, translating complex quantitative reasoning into clear, accessible explanations.
- Reviewed 13000+ peer-submitted solutions for mathematical accuracy and formatting consistency, functioning as subject-matter QA across a distributed contributor base.

ACADEMIC PROJECTS

Undergraduate Thesis Project | *April 2026*

Evaluating 14-Day Ahead Weather Predictions in the City of Manila: A Comparative Multivariate Study of ARIMAX and Lagged Multiple Linear Regression

- Applied Principal Component Analysis to 12 exogenous meteorological featuers, erducing dimensionality to 7 uncorrelated components used as model input.
- Developed ARIMAX and Lagged Multiple Linear Regression models in Python to generate 14-day ahead forercasts across three target labels.
- Evaluated model performance using RMSE, MAE, and other statistical tests, establishing ARIMAX's superiority in dampening error accumulation under high-frequency climate shocks.

TECHNICAL SKILLS

- **Machine Learning / Quantum Computing**: Qiskit, Variational Quantum Classifier, Quantum Image Encoding
- **Mathematics**: Linear Algebra, Probability, Statistics (Parametric/Non-parametric inference, Post-hoc analysis)
- **Programming/Tools**: Python (NumPy, Pandas, statsmodels), R, Microsoft Tools (Excel, Power BI), LaTeX (Overleaf, Texmaker), Basic SQL querying

CERTIFICATIONS and PROFESSIONAL DEVELOPMENT

Civil Service Professional Eligibility (94.60%) **Passed: May 2025**

Civil Service Commission